

Invited Paper: Multi-Agent Generative Synthesis for Analog/RF Circuit: from Scalable Topology Generation to Efficient Inverse Design

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Abstract—The exponential growth of information and computational workloads has created unprecedented demands for high-productivity development of computer hardware built on foundational semiconductor integrated circuits (ICs). Yet, the lack of effective design automation techniques makes developing analog ICs—indispensable in ubiquitous computer systems—a significant bottleneck for overall design productivity and cost efficiency across the IC ecosystem. Excitingly, recent advances in generative AI present transformative opportunities to tackle the complexity and large-scale challenges of modern analog/radio-frequency (RF) IC design. This work introduces a first-of-its-kind multi-agent generative synthesis framework for analog/RF circuits. Specifically, our approach formulates analog synthesis as a multi-stage generative AI problem: first generating a novel topology conditioned on high-level textual descriptions, and then producing high-quality device parameters to meet given design specifications for the generated circuit topology. This novel paradigm offers distinct advantages over traditional methods, including controllable novelty in topology generation and significantly improved inverse design efficiency. Beyond topology generation and inverse design, our method enables broader applications such as synthetic dataset generation and privacy-preserving data sharing—emerging challenges in data-driven electronic design automation (EDA) due to the computationally intensive and confidential nature of analog/RF circuit design. This work paves the way for next-generation generative AI-driven multi-agent synthesis in analog/RF EDA.

I. INTRODUCTION

Today's demand for the productivity and performance of foundational semiconductor integrated circuits (ICs) is soaring to unprecedented levels to meet the exponential growth of ongoing technological innovations and sustain future technological breakthroughs, such as generative AI, quantum computing, and more. While digital ICs have long enjoyed mature electronic design automation (EDA) flows for at-scale and efficient development, analog ICs, despite their essential role in ubiquitous electronic systems (e.g., conversion, transmission, and sensing), have historically relied on manual design processes due to their inherent complexities in abstraction [1], [2]. Such handcraft processes are notoriously time-consuming and costly, presenting a significant bottleneck to overall productivity, cost-effectiveness, and performance within the IC ecosystem. Over the past few decades, both academia and industry have devoted substantial effort to automating various tasks in analog IC design, leveraging traditional methodologies [3]–[7] as well as emerging AI-based approaches [8]–[15]. Despite notable progress, nearly all existing solutions remain narrowly specialized, often limited to specific circuit types and lacking generalizability across broader analog design spaces. This raises a critical question: what technological blueprint can overcome these limitations to enable generalizable analog EDA and catalyze the next leap in design productivity, performance, and innovation?

Recent breakthroughs in generative AI have empowered foundational models to perform a wide range of tasks at scale. These advances open new avenues for addressing the long-standing complex challenges of analog IC design, pushing beyond the constraints of traditional methodologies. In the domain of analog EDA, there is an increasing interest in harnessing generative AI to achieve generalizable design automation of analog ICs. Early pioneering work, CktGNN [10], uses a graph variational autoencoder (VAE) to produce circuit topologies for various operational amplifiers (Op-Amps). Recent work, LaMAGIC [11] fine-tunes a T5-small to generate diverse power converter topologies with varying device counts. AnalogCoder [12] has also shown that existing general large language models (LLMs) can be prompted to generate a broad spectrum of analog IC designs. More recently, domain-specific generative AI models, such as AnalogGenie [13], AnalogGenie-Lite [14] and AnalogFed [15], based on decoder-only transformers, have demonstrated exceptional capability in efficiently discovering and generating various analog circuit topologies while generalizing to different circuit types. This paradigm shift has shed light on a bright future of applying generative AI to achieve analog IC design automation in a general and scalable manner. Yet, despite these promising developments, much of this transformative potential remains untapped. A multi-agent generative AI framework capable of automating the entire analog IC design process—not just topology generation—represents a significant milestone. This end-to-end automation is the long-sought goal that could truly enable at-scale, next-generation AI-driven analog EDA.

As one key step towards this vision, this invited paper pioneers a generative AI-driven multi-agent design framework for analog/radio-frequency (RF) circuit synthesis. Our framework unifies novel topology discovery and efficient inverse design within a cohesive platform. A topology generation agent first produces candidate circuit structures based on high-level textual descriptions (e.g., “a two-stage op-amp with differential input pairs”). This agent builds upon our previous AnalogGenie model, now enhanced for controllability and specified generation. A performance extraction module then collects simulation data for the generated topology, which is used to train an inverse design agent that learns inverse mappings from circuit specifications to device parameters. Specifically, the inverse design agent is developed with a domain-specific conditional denoising diffusion probabilistic model (c-DDPM), which directly learns the reliable distribution of valid analog circuit physical device parameters given the target specifications of analog circuits as conditions. Our experiments find that generative inverse design offers several distinct advantages over

conventional methods. First, it enables the generation of diverse and valid design candidates in a single pass, unlike traditional approaches that typically yield only a single point estimate. Second, it shows superior data efficiency compared to reinforcement learning-based methods [16]–[18] due to its particular fit for sparse analog datasets. Last, it can generate high-quality synthetic data, which facilitates addressing the data scarcity challenge in AI-driven analog design due to privacy issues and accelerates downstream optimization by developing surrogate models. Together, these agents enable holistic analog/RF circuit synthesis from high-level descriptions to fully specified designs. Our work opens the door to efficient analog and RF design through a collaborative multi-agent framework based on generative AI. Key contributions are as below.

- **Novel generative synthesis framework for analog/RF ICs:** We introduce a first-of-its-kind multi-agent generative synthesis framework that harnesses generative AI for both topology generation and inverse design. This modular design enables holistic analog/RF circuit (active components) synthesis from high-level textual circuit descriptions to transistor-level parameters. The framework can simultaneously support multiple downstream tasks, offering broader applicability and greater flexibility than traditional methods.
- **Diverse, scalable, and specified topology generation:** We design an intelligent topology generation agent, which is capable of producing a wide variety of analog and RF circuit topologies with scalable parameters. Importantly, it also supports specification-guided generation, allowing users to define desired circuit types and guiding the model to generate topologies accordingly.
- **Efficient inverse design and high-quality synthetic data generation:** The inverse design agent, built on c-DDPM, can directly synthesize device parameters that meet specified performance targets and provides up to a $\sim 5\times$ speedup when used to warm-start conventional inverse design optimizers, significantly enhancing both search efficiency and success rates. It also produces synthetic datasets that support predictive models with high R^2 scores for benchmark circuits, closely matching real data performance. By addressing data scarcity and privacy concerns, this synthetic data generation capability also facilitates open-source data sharing in data-driven analog design automation.

II. BACKGROUND AND RELATED WORK

Analog circuit synthesis comprises two fundamental tasks: topology generation, which defines the structural configuration by interconnecting various devices, and inverse design (or device sizing), which determines the physical parameters required to meet target performance specifications. Recent advances in AI-driven EDA have explored addressing both tasks. Specifically, there is a notable increasing interest in leveraging generative AI models, such as Transformers, variational autoencoders (VAEs), and diffusion models, for topology generation. These models have shown great potential for discovering novel and diverse circuit structures. In this section, we begin by briefly reviewing the generative modeling techniques commonly adopted in analog design automation, with a particular focus on their application to topology generation. We then turn to the inverse design problem, where the use of generative models remains significantly underexplored but represents a promising direction for future research.

A. Generative Model Foundations

Generative AI models provide a powerful foundation for analog design automation, especially under inverse settings where the goal is to synthesize structural or parametric configurations that satisfy given targets. Representative modeling paradigms include:

1) *Transformers:* Auto-regressive transformer models [13], [14], [19] treat topology/parameter generation as a sequential generation task:

$$p(\mathbf{x}) = \prod_{i=1}^n p(x_i | x_{<i}), \quad (1)$$

where $\mathbf{x}$ represents a sequence of device tokens or sizing values. Transformers excel at modeling long-range dependencies and have been applied to analog topology generation tasks where the circuit structures are constrained by domain-specific rules and represented as token sequences (see AnalogGenie [13], AnalogGenie-Lite [14], and LaMAGIC [11]).

2) *Variational Autoencoders (VAEs):* VAEs [10] introduce a latent variable model $p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z})p(\mathbf{x}|\mathbf{z})$ and optimize the evidence lower bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q(\mathbf{z}|\mathbf{x})} [\log p(\mathbf{x}|\mathbf{z})] - D_{KL}(q(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})). \quad (2)$$

They are especially useful in capturing low-dimensional representations of circuit graphs or parameter spaces (See CktGNN [10]). Yet, compared to autoregressive transformers, VAEs often suffer from relatively low scalability and generalization capabilities.

3) *Diffusion Models:* Denoising diffusion probabilistic models (DDPMs) [20] generate data by learning to reverse a Gaussian corruption process:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (3)$$

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)). \quad (4)$$

Diffusion models are particularly effective for high-fidelity sample generation and can be adapted for conditional inverse design (see Diffckt [21]).

Recognizing that different generative methods are better aligned with different sub-tasks, we design a multi-agent generative synthesis framework that applies transformers to topology generation and diffusion models to device sizing.

B. Analog Circuit Topology Generation

The topology design requires the most creative effort in analog IC design, as it is conventionally conceptualized from scratch by human designers. This task explores a large and discrete design space, making it structurally complex and critical to circuit functionality and performance potential. Historically, progress in analog circuit performance has been primarily driven by semiconductor process scaling. Yet, with the end of Moore's Law on the horizon, continued performance improvements can no longer benefit from technology scaling alone. Developing novel analog circuit topologies has arisen as a promising pathway to sustain and advance analog IC performance in the post-Moore era, much like developing new protein structures for the generation of improved amino acid sequences. Given its importance, generative AI has been increasingly used in topology design/discovery and has become the major methods.

One of the earliest works, CktGNN [10], formulates topology design as a graph generation task and uses a variational autoencoder to produce various operational amplifier (Op-Amp) topologies. LaMAGIC [11] introduces a masked language model to generate power converter designs with varying devices. AnalogCoder [12] applies

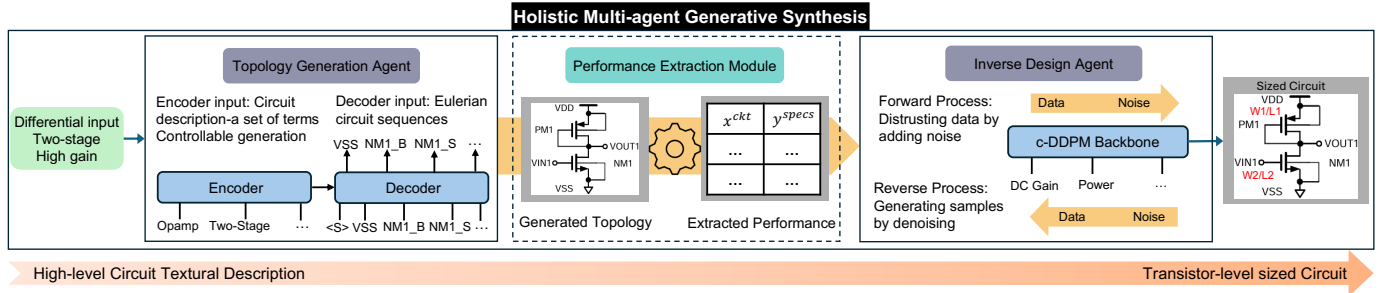


Fig. 1: Overview of the proposed multi-agent generative synthesis framework, which integrates topology generation, performance extraction, and inverse design to enable end-to-end analog circuit design from high-level textural description to transistor-level sized circuits.

domain-specific prompting engineering to existing LLMs to generate PySpice code, which is then converted into circuit topologies. AnalogGenie [13] and its lightweight variant AnalogGenie-lite [13], build on decoder-only transformers, adopt a scalable, sequence-based graph representation and demonstrate state-of-the-art performance in generating diverse, complex, and previously unseen topologies. AnalogGenie-Lite [14] makes analog circuit topology discovery both more efficient and broadly applicable by further optimizing the underlying graph representation of AnalogGenie.

While these methods demonstrate the great promise of generative AI in analog topology generation, several key limitations hinder their integration into holistic design workflows. First, most approaches treat topology generation as an isolated task (with the exception of LaMAGIC [11]), lacking integration with downstream processes such as performance evaluation or inverse design. Second, their generation quality and generalization capabilities remain limited. For instance, AnalogCoder [12] demonstrates that even state-of-the-art LLMs often fail to generate valid analog circuits within five attempts, and LaMAGIC [11] is narrowly tailored to power converter topologies. Third, many methods, including AnalogGenie [13] and AnalogGenie-Lite [14], lack controllable conditioning mechanisms, making it difficult to steer generation toward specific circuit types. To overcome these challenges, we propose an enhanced topology generation agent built on AnalogGenie [13]. This upgraded agent supports precise, scalable, and controllable generation of analog and RF circuit topologies, while also interfacing seamlessly with downstream performance evaluation and inverse design agents—enabling holistic analog circuit synthesis.

C. Analog Circuit Inverse Design

Inverse design, also known as device sizing, is the complementary task to topology generation. Once a topology is created, inverse design is performed to determine physical parameters, such as transistor widths and lengths, to meet specific performance targets (e.g., gain, bandwidth, noise, power). Traditionally, inverse design has been studied with optimization techniques such as Bayesian optimization [22], geometric programming [23], genetic algorithms [3], and simulated annealing [24]. While effective in small-scale design problems, these approaches are often sensitive to initialization, and struggle with scalability [25]. Learning-based approaches, especially reinforcement learning (RL) [16], [26], have emerged as promising alternatives for large-scale design. Yet, they typically require time-consuming simulator-in-the-loop training. To mitigate this, surrogate models have been developed to replace circuit simulators [27]–[30], improving training efficiency. Hybrid approaches, such as Rose [18] and Rose-OPT [31], combining BO for coarse exploration with RL for fine-tuning, also yield improved data efficiency and final performance. More recently, LLM-based methods have shown promise

in enhancing inverse design efficiency. ADO-LLM [32] integrates LLMs with BO to improve inverse design efficiency through in-context learning, while LEDRO [25] employs LLMs to effectively reduce the device parameter search space, facilitating more efficient downstream optimization.

Despite recent progress, inverse design remains a simulation-bound and data-intensive process. Specifically, developing accurate surrogate models also require extensive and time-consuming circuit simulations to construct reliable ground-truth datasets. Compounding this challenge, these datasets are often derived from commercial semiconductor technologies protected by intellectual property and confidentiality constraints, making open-source sharing infeasible. This lack of accessible data hinders reproducible research, fair benchmarking, and broader adoption of design tools within the research community. These challenges are becoming increasingly severe, underscoring the urgent need for innovative methods that can simultaneously ensure design quality, improve efficiency, and preserve data privacy. We introduce a novel generative inverse design agent that can meet all these needs and serves as a core component of our proposed multi-agent generative synthesis framework. It can form a hybrid or combined optimization strategy that balances performance and efficiency by seamlessly integrating with other methods. It can also enable open-source data sharing by generating high-fidelity synthetic datasets, which can support broader accessibility and collaboration in AI/ML, EDA, and IC communities.

III. MULTI-AGENT GENERATIVE SYNTHESIS FRAMEWORK

A. Framework Overview

Our multi-agent generative synthesis framework integrates two specialized agents and one modular component to enable a holistic and scalable pipeline for analog and RF circuit synthesis (Fig. 1): a topology generation agent, a performance extraction module, and an inverse design agent. Given a high-level textual description (e.g., “a two-stage Op-Amp with differential input pairs”), the topology generation agent generates candidate circuit graphs conditioned on the target functionality and structural constraints. The generated circuits are then evaluated by the performance extraction module, which simulates each topology to extract key performance metrics (e.g., gain, bandwidth, phase margin), thereby constructing a dataset of paired device-level parameters and corresponding specifications. This dataset is used to train the inverse design agent, which learns a conditional generative model over the device parameter space given target specifications. Once trained, the agent can synthesize valid transistor-level parameters directly from new specification queries, thus enabling efficient and specification-driven inverse design.

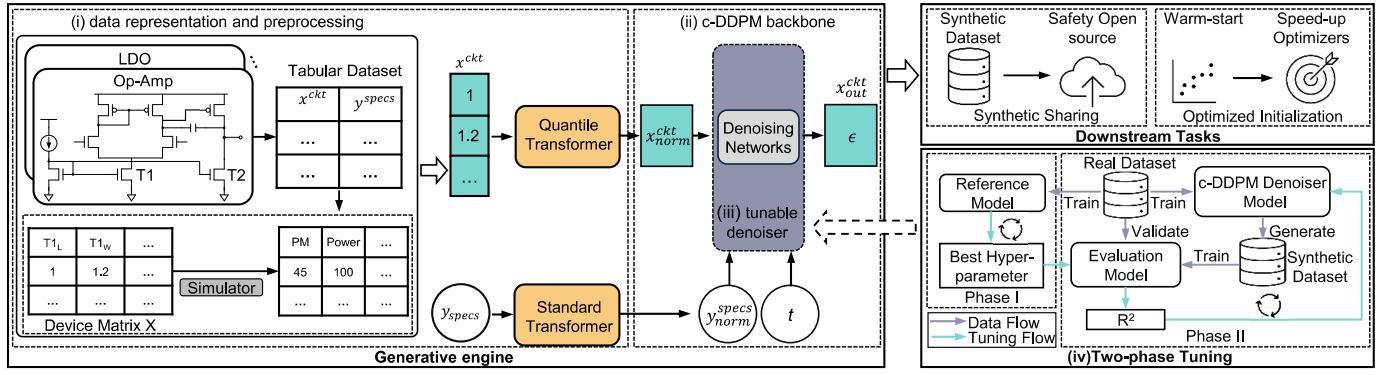


Fig. 2: Inverse design agent begins with the schematic-to-tabular transformation (i), where fixed-topology analog circuits are mapped into a structured tabular dataset of device parameters and corresponding specifications from Performance Analysis Agent. Circuit parameters ($\mathbf{x}^{\text{ckt}}$) are normalized using a quantile transformer, and specifications ($\mathbf{y}^{\text{specs}}$) are standardized using a standard scaler. The embedded specification vector and diffusion time step (t) are fed into the c-DDPM backbone (ii) to predict the added noise (ϵ) during the forward process. The generated data can be applied to warm-starting circuit sizing optimizers or high-fidelity synthetic dataset release. Illustration of the two-phase tuning framework (iv) designed to optimize the denoising network (iii): the reference model tuning phase (Phase I) and the c-DDPM denoiser model tuning phase (Phase II).

B. Topology Generation Agent

To enable controllable and specification-driven analog/RF circuit topology generation, we propose an encoder-decoder architecture for the topology generation agent, extending our earlier work on AnalogGenie, a decoder-only Transformer model. In this enhanced design, the encoder processes a tokenized, natural language-style description of the target circuit (e.g., “two-stage op-amp”, “high gain”, “differential input”), capturing the desired functional and structural attributes. The decoder is trained to generate the corresponding Eulerian circuit sequence, representing a traversable path through the circuit graph, consistent with the sequence format used in AnalogGenie. During training, we employ teacher forcing by feeding a shifted version of the target sequence into the decoder to stabilize and accelerate learning. This encoder-decoder formulation is well-suited for practical design scenarios where users require specific circuit types with targeted behaviors. Compared to the original AnalogGenie, which lacks input conditioning and generates random topologies, our approach provides significantly greater control over the generation process and yields outputs that are more interpretable, design-aligned, and usable in real-world applications. We omit the detailed technical description of this enhanced generative model, as its core architecture and training procedure closely follow those of AnalogGenie. Our focus is on the inverse design agent.

C. Inverse Design Agent

We also develop a domain-specific generative AI agent to address inverse design. Its cornerstone is a customized generative engine for high-quality production of physical device parameters, which can be used to enable several downstream sub-tasks related to the inverse design (Fig. 2). To achieve the superb capability of the engine, we tailor it with four key elements: (i) data representation and preprocessing, (ii) a conditional DDPM (c-DDPM) backbone, (iii) a tunable denoiser for score estimation and conditional embedding, and (iv) two-phase tuning for the best generation quality. With these elements and a ground-truth dataset consisting of sparse samples of circuit specifications and device parameters in the design space, the engine learns to generate a distribution of corresponding design parameters that are highly likely to satisfy the corresponding specifications. Then it can be applied to enable several downstream tasks.

1) *Tailored Data Representation and Preprocessing*: As shown in Figure 2(i), in the inverse design of analog circuits with a fixed topology, each device (e.g., a transistor) is characterized by a small set of physical parameters—such as length, width, and multiplier—that can be easily modeled as a short vector. By concatenating the parameter vectors of all devices, we represent the complete set of device parameters as a single, long vector. Similarly, circuit performance specifications (e.g., gain, phase margin, and bandwidth for an Op-Amp) are also represented as a vector, which serves as the conditional input to our generative engine. Here, the circuit topology and the associated (parameter, specification) pairs are provided by the topology generation agent and the performance analysis agent, respectively. Importantly, while there is a deterministic one-to-one mapping from a specific device parameter vector to its corresponding circuit specifications, the inverse mapping is inherently one-to-many and non-convex: multiple combinations of device parameters can yield the same performance metrics. This fundamental asymmetry renders conventional optimization and RL methods that often produce a single group of optimized device parameters sub-optimal. With this data representation in mind, we create a ground-truth tabular dataset for each circuit, where each row contains a device parameter vector $\mathbf{x}^{\text{ckt}}$ and its corresponding specification vector $\mathbf{y}^{\text{specs}}$. The conditional generation task then becomes a matter of learning the distribution $p(\mathbf{x}^{\text{ckt}} | \mathbf{y}^{\text{specs}})$, for which diffusion models offer a natural and powerful solution due to their ability to model complex, multimodal distributions. This dataset is further preprocessed for use by the backbone c-DDPM. Specifically, the device parameters $\mathbf{x}^{\text{ckt}}$ are normalized using a quantile transformer to preserve marginal distributions, while specification vectors $\mathbf{y}^{\text{specs}}$ are standardized to stabilize conditioning.

2) *Customized c-DDPM Backbone*: To enable the conditional generation process shown in Figure 2(ii), we customize a domain-specific c-DDPM to operate on the proposed vectorized circuit representation. The objective of c-DDPM is to learn the conditional distribution $p(\mathbf{x}_0^{\text{ckt}} | \mathbf{y}^{\text{specs}})$, where $\mathbf{x}_0^{\text{ckt}}$ denotes the design parameters of an analog circuit (e.g., transistor widths, lengths, and multiplicities), and $\mathbf{y}^{\text{specs}}$ represents the corresponding performance specifications (e.g., gain, bandwidth, phase margin, etc.). The goal is to generate design parameters that satisfy the given specification constraints. We develop

this c-DDPM for vectorized circuit representation by following the essential forward process, reverse process, and training objective of implementing a general diffusion model [20]. During training, Gaussian noise is added to the design parameters according to a fixed forward noise schedule. The denoising network ϵ_θ takes as input the noisy parameters, the time step t , and the embedded specification $\mathbf{y}_{\text{specs}}$, and learns to predict the added noise. Once trained, at the generation phase, a valid set of circuit device parameter vector that satisfies the given circuit specification is generated by initializing with Gaussian noise $\mathbf{x}_T^{\text{ckt}} \sim \mathcal{N}(0, \mathbf{I})$ and iteratively applying the learned reverse dynamics:

$$\mathbf{x}_{t-1}^{\text{ckt}} = \mathbf{x}_t^{\text{ckt}} - f_\theta(\mathbf{x}_t^{\text{ckt}}, t, \mathbf{y}_{\text{specs}}) + \text{noise}, \quad (5)$$

where f_θ is derived from the estimated score. The final sample $\mathbf{x}_0^{\text{ckt}}$ represents a set of design parameters expected to meet the given circuit specification $\mathbf{y}_{\text{specs}}$.

3) *Denoising Network and Conditional Embedding*: We adopt three structures for denoising networks: MLP, ResNet, and Transformer, for sufficient explorations (Figure 2(iii)). The primary objective is to determine whether different network structures can significantly impact the performance of the c-DDPM. All models take as input a noisy circuit parameter vector $\mathbf{x}_t^{\text{ckt}}$, a diffusion timestep t , and the target circuit specifications $\mathbf{y}_{\text{specs}}$, and output a noise estimate $\hat{\epsilon}$. While these three networks have different structures, we develop the same time step and specification embedding for them. Specifically, the diffusion timestep t is processed with a sinusoidal positional encoding followed by two linear layers and a SiLU nonlinearity:

$$\mathbf{t}_{\text{emb}} = \text{Linear}(\text{SiLU}(\text{Linear}(\text{SinTimeEmb}(t)))). \quad (6)$$

For the multi-dimensional specification vector $\mathbf{y}_{\text{specs}} \in \mathbf{R}^d$, we use a structured embedding strategy:

$$\mathbf{y}_{\text{emb}} = \text{SiLU} \left(\sum_{i=1}^d \text{Linear}_i(y_i) \right), \quad (7)$$

where each y_i is embedded individually and summed to form a shared conditioning vector. This modular design ensures independent control across specification dimensions while preserving a unified representation.

Hierarchical Transformer with Local and Global Attention. While the development of MLP and ResNet-based denoising networks is relatively straightforward, we focus on a domain-specific Transformer architecture tailored for analog circuit data. This design incorporates analog-specific inductive biases through hierarchical attention mechanisms to enhance denoising performance. To effectively capture structural priors in analog circuit parameter spaces, we implement a hierarchical Transformer denoiser that stacks N encoder blocks on top of an input embedding. Each encoder block consists of alternating local and global multi-head attention (MHA) layers with residual connections. The core computation within each encoder block is defined as:

$$\text{Encoder}_i(\mathbf{x}) = \mathbf{x} + \text{FFN}(\mathbf{x} + \text{MHA}_i(\mathbf{x})), \quad (8)$$

$$\text{FFN}(\mathbf{x}) = \text{Linear}(\text{ReLU}(\text{Linear}(\mathbf{x}))). \quad (9)$$

$$\text{MHA}_i(\mathbf{x}) = \begin{cases} \text{Local-MHA}(\mathbf{x}), & \text{if } i \bmod 2 = 1, \\ \text{Global-MHA}(\mathbf{x}), & \text{otherwise,} \end{cases} \quad (10)$$

The local MHA restricts attention within parameters of the same device (e.g., width W and length L), while the global MHA enables cross-device interactions, capturing inter-device coordination patterns

crucial in analog design. This hierarchical design embeds circuit-aware inductive priors and shows superior performance in both generation quality and downstream optimization tasks (see evaluation part).

4) *Two-Phase Tuning for High Generation Quality*: The generation quality of c-DDPM can be sensitive to hyperparameters and network structure parameter choices, which need to be evaluated and tuned. Unlike conventional supervised learning models that can be directly evaluated using ground-truth targets, generative models lack explicit reference outputs, especially under the context of circuit design. One straightforward approach is to perform circuit-level simulations using the generated device parameters to obtain the actual circuit specifications and compare these ground-truth specifications with the given targets. However, circuit-level simulations are extremely time-consuming, making this way inefficient. As a result, their quality must be assessed indirectly—typically by assessing how well a downstream evaluation model (i.e., forward model), trained on the generated data, can approximate the real data distribution. This indirect evaluation highlights a key challenge: without a direct performance reference, tuning the c-DDPM requires a careful and principled approach. To address this issue, we propose a two-phase tuning strategy, illustrated in Figure 2(iv), that decouples the optimization of the evaluation model (Phase I) from that of the generative model (Phase II). In Phase I, we tune a supervised regression model on the real dataset to establish a reliable performance baseline, i.e., maximizing the R^2 score on the real-world dataset. Once the best-performing configuration is achieved, the identified parameters are fixed and reused to train an evaluation model in Phase II. The evaluation model will be trained on synthetic data generated by c-DDPM and will serve as a consistent evaluator to indirectly assess the quality of the generated data. In Phase II, we tune the denoising network of c-DDPM to achieve the best generation quality. During tuning, the architecture parameters and hyperparameters of the evaluation model remain fixed. We instead explore different denoiser designs—either MLPs, ResNets, or Transformers—and tune their depth, hidden dimensions, and training parameters. A Bayesian optimization strategy [33] is employed to automate this tuning process by identifying the denoiser design within a predefined step. The best denoiser is the one that can maximize the R^2 achieved by the evaluation model. This is because a well-trained surrogate not only captures the marginal distribution alignment between synthetic and real data, but also preserves meaningful relationships among internal features and specifications, making its R^2 a practical and principled merit for tuning generative quality. This indirect yet performance-driven tuning loop enables us to systematically discover the most effective generative model, without requiring time-consuming cross-validation with circuit simulators.

5) *Downstream Tasks*: Upon tuning, this optimized generative engine supports two primary downstream tasks. First, it can produce high-quality initial solutions that either directly satisfy the design goals or fall into near-feasible/high-performing regions where optimal solutions typically. As such, even in the worst case, it will effectively serve as a sub-optimal optimizer and enable seamless integration with existing device-sizing methods (i.e., GA, BO, and RL) to form a hybrid or combined optimization strategy, significantly enhancing design efficiency and increasing success rates by providing optimized warm-start points. Second, the generative engine inherently supports open and privacy-preserving data sharing by generating abundant high-fidelity synthetic specification/device-parameter pairs that closely match the distribution of real data. This unique capability facilitates collaborative innovation while protecting proprietary infor-

TABLE I: Overview of our dataset. It includes a total of 3,350 unique circuits spanning both traditional and RF active categories.

Circuit Type	Count	Category
Op-Amp	836	Analog
LDO	321	Analog
Bandgap Reference	667	Analog
Comparator	87	Analog
PLL	10	Mixed-Signal
LNA	47	RF Active
PA	27	RF Active
Mixer	46	RF Active
VCO	233	RF Active
Power Converter	353	Analog
Switched-Capacitor Sampler	723	Mixed-Signal
Total	3,350	

TABLE II: Qualitative comparison between AnalogGenie and our encoder-decoder-based model.

Aspect	AnalogGenie	Ours
Architecture	Decoder-only	Encoder-Decoder
Input Description	Not supported	Supported
Generation	Random topology	Conditioned
Training Supervision	Sequence-only	Description $\rightarrow$ Sequence
Interpretability	Low	High (description-aligned)
Design Space	Unstructured	Targeted and diverse
Validity	High	High
Scalability	Medium	Medium
Novelty	High (randomized)	Moderate (goal-driven)

mation, advancing the broader goal of accessible and reproducible analog design automation research, surpassing the capability of any existing method. The experimental evaluations of these two tasks are presented in the next Section.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

Circuit benchmark: For topology generation evaluation, we leverage the AnalogGenie dataset containing 3,350 unique topologies across 11 circuit types, including Op-Amps, LDOs, PLLs, and RF active circuits such as mixers, LNAs, and PAs, as shown in Table I. Each circuit is annotated with a descriptive label (e.g., differential input, two-stage, high gain). During training, the encoder is provided with these natural language-style descriptions, while the decoder is supervised to generate the corresponding Eulerian circuit sequence, enabling description-driven topology generation. We evaluate the inverse design agent on both analog and RF circuits: a two-stage Op-Amp and an LDO using the open-source SkyWater 130-nm PDK [34] with open-source simulation toolchain (e.g., Ngspice, Xyce), and a digitally controlled oscillator (DCO, a subclass of voltage-controlled oscillators, VCOs) using the proprietary GlobalFoundries 22-nm CMOS SOI PDK. Following [35], [36], we consider four key specifications for the Op-Amp: DC gain, GBW, phase margin (PM), and power; for the LDO: line/load regulation (LNR/LDR), PM, and offset voltage; and for the DCO [37]: oscillation frequency, phase noise, and power. Analog circuits typically have tens of free parameters, including transistor sizing, capacitor geometry, and bias currents, whereas the RF circuit has five physical parameters. Yet, the framework can be generalized to various types of analog/RF circuits.

Evaluation strategy: We use the same metric proposed by AnalogGenie to evaluate topology generation quality. We evaluate the generation quality of the inverse design agent using both model-based and simulation-based strategies. Beyond generation quality, we assess its utility in two downstream tasks: warm-starting analog circuit optimizers and constructing high-fidelity, privacy-preserving

TABLE III: Regression performance (R^2) of Transformer models trained on real versus inverse design agent-generated data. We evaluate both $1\times$ and $10\times$ synthetic data scales, using MLP, ResNet, and Transformer denoisers. All models are tested on real data for Op-Amp and LDO circuits.

Setting (Training $\rightarrow$ Test)	Op-Amp		LDO	
	Val R^2	Test R^2	Val R^2	Test R^2
Real $\rightarrow$ Real (Exp A)	0.8763	0.8667	0.9586	0.9657
Synth. (MLP, Exp B) $\times 10$	0.8919	0.8803	0.9120	0.9384
Synth. (MLP, Exp B) $\times 1$	0.8144	0.7999	0.8326	0.8500
Synth. (ResNet, Exp B) $\times 10$	0.8622	0.8496	0.9305	0.9488
Synth. (ResNet, Exp B) $\times 1$	0.8032	0.7908	0.8428	0.8819
Synth. (Transformer, Exp B) $\times 10$	0.9020	0.8938	0.9345	0.9491
Synth. (Transformer, Exp B) $\times 1$	0.8030	0.7966	0.834	0.8900
Rand. Init (MLP, Ablation) $\times 10$	0.8119	0.7995	0.8846	0.8912
Rand. Init (ResNet, Ablation) $\times 10$	0.8467	0.8287	0.9152	0.9232

TABLE IV: Regression performance (R^2) of Transformer models trained on real versus inverse design agent-generated data. Evaluated on DCO circuit.

Setting (Training $\rightarrow$ Test)	Val R^2	Test R^2
Real $\rightarrow$ Real (Exp A)	0.9271	0.9307
Synth. (MLP, Exp B) $\times 10$	0.9627	0.9685
Synth. (ResNet, Exp B) $\times 10$	0.9704	0.9694

datasets for open-source release. **Dataset and training:** We split each dataset (15K Op-Amp, 10K LDO, 21K DCO) into 80%/10%/10% train/val/test. The c-DDPM is trained on real data and generates synthetic samples to train a Transformer regressor, evaluated on held-out real data. For completeness, we include brief performance summaries of the topology generation agent, but omit detailed setups, as its model architecture, training procedure, and evaluation protocols are directly adapted from AnalogGenie, and no major experimental changes were introduced. Our primary focus in this work is on the inverse design agent.

B. Experimental Results

1) *Topology Generation:* Table II summarizes the key differences between AnalogGenie and our enhanced encoder-decoder-based model. While both models exhibit comparable levels of validity and scalability, our model offers significant advantages in controllability, interpretability, and design-space alignment. Unlike AnalogGenie, which is a decoder-only, sequence-only generator that produces random topologies without leveraging design intent, our model utilizes structured circuit descriptions through an encoder. This transition from unconditional to conditional generation enables outputs that are better aligned with user-specified requirements. Although this added guidance may slightly reduce generation novelty, it significantly improves the relevance of generated circuits, making the model more practical and effective for real-world analog design applications.

2) *Model-Based Evaluation for Synthetic Data Generation Quality and Utility:* We first assess the quality and utility of the synthetic data generated by inverse design agent using a model-based method. Specifically, we conduct two regression experiments using a Transformer-based forward model on each circuit. In Experiment A (Exp A), we train a model on real-world data and evaluate on the real-world test dataset. This serves as a performance upper bound, providing a benchmark for downstream prediction accuracy. In Experiment B (Exp B), we train an identical model using only synthetic data generated by inverse design agent and evaluate it on the same real-world test dataset. This setup reflects a practical scenario

TABLE V: Wasserstein Distance (WD) for different denoiser architectures on Op-Amp and LDO circuits.

WD (avg.) ↓	MLP	ResNet	Transformer
Op-Amp	0.0251	0.8763	0.0250
LDO	0.0433	0.0288	0.0286

TABLE VI: Simulation-based validation of inverse design agent-generated inverse designs. Each subtable reports the MAE and RMAE (%) between simulated circuit performance and target specifications.

(a) Op-Amp			(b) LDO		
Spec	MAE	RMAE	Spec	MAE	RMAE
DC gain (dB)	1.36	1.56	VOS (V)	0.0006	4.59
Power (μ W)	7.55	4.57	LNR (1/V)	0.0269	9.08
PM ($^\circ$)	5.13	1.43 [†]	LDR (1/A)	0.0029	4.20
GBW (MHz)	0.20	11.08	PM ($^\circ$)	1.27	0.35 [†]

[†]RMAE normalized against 360 $^\circ$ margin.

where synthetic data is used for surrogate modeling when access to the original data is restricted due to privacy or proprietary concerns. Table III reports the R^2 scores on both validation and test sets for analog circuits (Op-Amp and LDO). We evaluate three variants of inverse design agent, employing MLP, ResNet, and Transformer as the denoising network. While overall performance differences among denoising architectures are moderate, our proposed domain-aware Transformer consistently achieves the highest R^2 scores across both Op-Amp and LDO tasks, highlighting its effectiveness in modeling analog circuit priors and guiding inverse design. To further validate the effectiveness of our two-phase tuning strategy, we conduct an ablation study using randomly initialized denoisers without tuning. As shown in the last two rows of Table III, these ablation results yield noticeably lower R^2 scores, confirming the importance of our tuning process in enhancing generation quality. We additionally report the R^2 scores on both validation and test sets for the RF circuit benchmark (i.e., DCO), as shown in Table IV. For this task, we evaluate inverse design agents using MLP and ResNet denoising networks. Consistent with the results on analog circuits, the high R^2 scores achieved in Experiment B demonstrate that our inverse design agent generalizes effectively to RF circuit synthesis, maintaining strong generation quality across different circuit domains.

We also analyze how the quantity of synthetic samples affects downstream regression. For both Op-Amp and LDO, forward models trained on $10\times$ synthetic data achieve significantly higher R^2 on real test sets compared to those trained on $1\times$ synthetic data (equal in size to the real dataset). This highlights the utility of synthetic data: while collecting $10\times$ real samples via simulation would be prohibitively expensive, inverse design agent enables cost-effective data scaling. Additionally, our domain-aware Transformer denoiser consistently achieves higher R^2 and lower WD across tasks (Table V), confirming its effectiveness and the benefit of our tuning strategy.

3) *Simulator-in-the-Loop Evaluation for Effectiveness of Inverse Design*: We also perform simulation-in-the-loop validation to assess the practical effectiveness of inverse designs generated by inverse design agent. Specifically, we randomly sample 10 sets of target specifications for each circuit type (Op-Amp and LDO). For each target, inverse design agent generates 20 sets of device parameters conditioned on the given specifications. These generated designs are then evaluated through circuit-level simulations using the SkyWater 130-nm PDK to obtain ground-truth performance metrics. Table VI summarizes the errors between circuit simulation results (based on the

TABLE VII: Optimization steps required to reach target specifications using GA and RL under different initialization strategies. Inverse design agent significantly accelerates convergence.

Task	Rand. Init	Agent Init	Gain ↑	Data ↓
GA on Op-Amp	>100	25	> 4.00	–
GA on LDO	14	3	4.67	–
RL on Op-Amp	27	5	5.40	140k
Inverse Design Agent	–	1	–	<15k

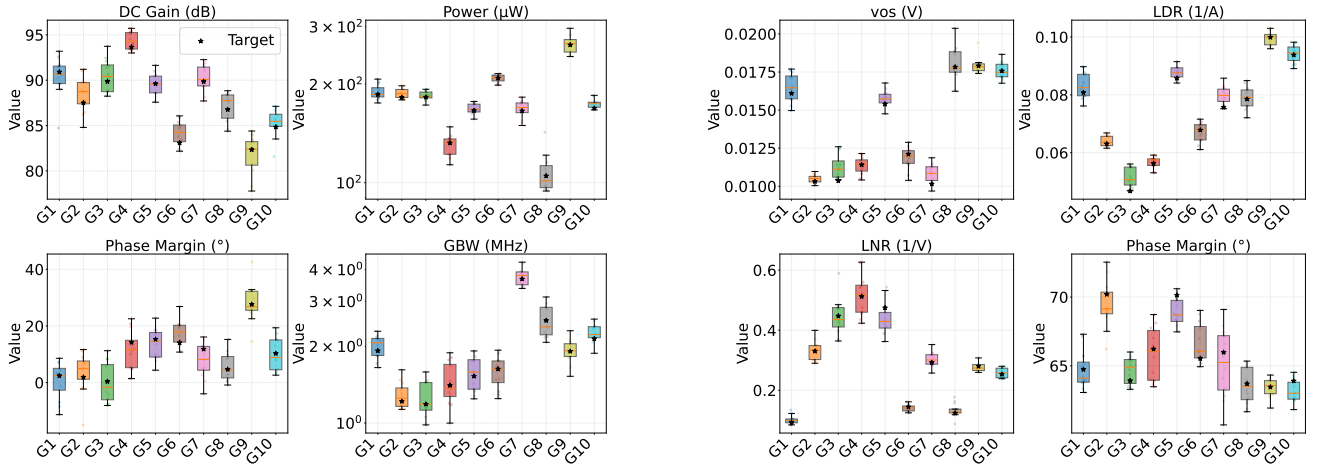
generated inverse designs) and the corresponding target specifications (i.e., conditions) across all inverse design agent-generated samples. We report the mean absolute error (MAE) and the relative mean absolute error (RMAE) for each specification to assess performance quantitatively.

To provide a more intuitive view of the deviation between target and simulated values, Figure 3 illustrates the distribution of relative errors using quartile-based boxplots. These results highlight the strong consistency of inverse design agent’s conditional generation across a diverse set of specifications and circuit types. Overall, the findings demonstrate that inverse design agent reliably produces design candidates whose simulated performance closely matches the intended targets, validating its effectiveness for closed-loop analog circuit design. These candidates can be directly used as optimal design parameters to meet the design goals or, in the worst case, as warm-starting points in a hybrid/combined optimization method for further optimization, as shown below.

4) *Downstream Task Evaluation for Inverse Design Agent*: We further evaluate the practical utility of the proposed inverse design agent across two downstream tasks that reflect realistic usage scenarios in analog/RF design.

Task-1: Warm-Starting Downstream Optimizers. The generation capability (i.e., probabilistic generation) of inverse design agent makes it especially well-suited for warm-starting downstream optimization algorithms, such as GA or RL, by providing high-quality initial candidates that lie closer to feasible or high-performing regions in the design space. As a result, inverse design agent can significantly accelerate convergence and improve overall design efficiency. We evaluate this capability using both GA and RL optimization flows, implemented with the SkyWater 130 nm PDK, where each candidate design is validated through circuit simulations. For GA, we fix the population size to 50 and continue optimization until the best candidate meets all target specifications.

Table VII reports the number of optimization steps required to reach convergence under two initialization strategies: random initialization (init) and inverse design agent-based initialization (init). The results show that inverse design agent dramatically reduces the number of optimization steps. For example, in the Op-Amp design task, GA fails to converge within 100 generations when initialized randomly, while inverse design agent enables convergence within just 25 generations. Similarly, for LDOs, GA converges in 3 generations under inverse design agent initialization compared to 14 generations under random initialization. We also observe a similar benefit in RL-based optimization for Op-Amps: convergence is reached in only 5 steps with inverse design agent initialization, compared to 27 steps with random initialization. In the case that inverse design agent can directly generate the desired device parameters, there is only one step needed, indicating a much higher design efficiency. It is also important to note that RL requires 140,000 circuit-level simulations to achieve convergence, as reported in the original work [16], while inverse design agent only requires 15,000 circuit-level simulations,



(a) Op-amp: distribution of simulated specification values across 10 target groups.

(b) LDO: distribution of simulated specification values across 10 target groups.

Fig. 3: Boxplot comparison between target specifications and simulated performance of inverse design agent-generated designs across 10 specification groups for (a) Op-amp and (b) LDO circuits. Each subplot corresponds to one specification and shows the distribution of simulated values in their original physical units (e.g., dB, V, °, MHz) across 20 generated samples per group. Target specification values are indicated by star markers (shown as black stars).

TABLE VIII: Privacy and distributional similarity metrics comparing inverse design agent-generated data with real datasets.

Metric	Op-Amp	LDO
WD (avg.)	0.0251	0.0288
DCR (avg.)	0.9277	0.3625
DCR (min)	0.4507	0.0501
Label corr. (Pearson)	0.9387	0.9865
Mean shift (%)	4.35%	2.89%
Std dev. shift (%)	2.44%	6.22%

representing an almost $10\times$ data efficiency improvement. These results highlight inverse design agent’s ability to identify high-quality regions in the design space and to effectively guide both evolutionary and policy-based optimization processes, demonstrating its practical utility in real-world analog design automation workflows. While our experiments focus on GA and RL, we note that inverse design agent-generated warm-start candidates are also applicable to other black-box optimizers such as BO. Since GA and BO both rely on iterative evaluation and sampling from the design space, the performance gain observed with GA warm-starting is expected to extend to BO as well.

Task-2: Privacy Evaluation. Finally, we evaluate the privacy-utility trade-off of inverse design agent-generated data using two metrics [38]: Wasserstein Distance (WD) for global distribution alignment, and Distance to Closest Record (DCR) for individual-level privacy. As shown in Table VIII, inverse design agent achieves low WD (0.0251 for Op-Amp, 0.0288 for LDO), confirming close alignment with the real data. The minimum DCRs (0.4507 and 0.0501) ensure no synthetic sample directly replicates any training point. LDO’s lower DCR is due to its denser design space but does not indicate memorization. Label correlations (0.9387 and 0.9865) and $<6.5\%$ mean/std shift of the full dataset confirm strong statistical fidelity. These results show that inverse design agent preserves privacy while maintaining high utility, making it safe for release even with proprietary PDK data. The synthetic data can be broadly used to fine-tune surrogate (forward) models through transfer learning, further improving the efficiency of current analog synthesis methods, such

as GA, BO, and RL.

C. Summarization and Discussion

Overall, our experiments demonstrate that the proposed multi-agent generative synthesis framework successfully unifies topology generation and inverse design into a holistic framework. The topology generation agent enables controllable and description-guided design, while the inverse design agent consistently produces high-quality parameter sets that meet target specifications under simulation. Additionally, it improves the efficiency of downstream optimization and facilitates privacy-preserving synthetic data generation. These results underscore the strong potential of generative AI to drive the next generation of scalable, modular, and intelligent analog IC design.

Despite the promising performance, we would like to discuss several knobs for further improvement. **Generalizable inverse design:** The current inverse design method is still circuit-specific. We will explore enhanced techniques to make it generalizable to different circuits. **Variation adaptability:** While the inverse design agent demonstrates strong generalization to unseen specifications, it does not yet account for process-voltage-temperature (PVT) variations—an essential capability for robust and practical deployment in real-world design environments. In the future, we also plan to explore tighter agent coordination to enable interactive refinement between topology and sizing agents.

V. CONCLUSION

We propose a multi-agent generative synthesis framework for end-to-end analog/RF front-end circuit design, capable of transforming high-level textual circuit descriptions into transistor-level parameters. Our extensive experiments validate its effectiveness and demonstrate its capability to support multiple tasks: controllable topology generation, specification-driven inverse design, optimization warm-starting, and privacy-preserving data sharing. Overall, the framework offers a unified solution for analog circuit synthesis, performance optimization, and secure data utilization, paving the way for scalable and flexible design automation. Acknowledgment: This work is partly supported by the Natcast AIDRFIC project.

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